Resources and Development

Class 10 Geography — Detailed NCERT Notes

1. Concept of Resources

Everything available in our environment which can be used to satisfy our needs is termed as a 'Resource', provided it meets three critical criteria: it must be **technologically accessible**, **economically feasible**, and **culturally acceptable**.

Resources are not free gifts of nature; they are a function of human activities. Human beings interact with nature through technology and create institutions to accelerate economic development.

- **The Transformation Process:** Involves an interactive relationship between **Nature** (Physical Environment), **Technology**, and **Institutions**.
- **Human Role:** Humans are essential components of resources because they transform material available in the environment into usable resources.

2. Classification of Resources

Resources are classified into various categories based on different criteria:

- **On the basis of Origin:** Biotic (living, e.g., flora, fauna) and Abiotic (non-living, e.g., rocks, metals).
- **On the basis of Exhaustibility:** Renewable (can be reproduced, e.g., solar, wind, forests) and Non-renewable (take millions of years to form, e.g., fossil fuels, minerals).
- **On the basis of Ownership:** Individual, Community, National, and International resources.
- **On the basis of Status of Development:** Potential, Developed Stock, and Reserves.

3. Development of Resources and Sustainability

Indiscriminate use of resources by humans has led to major global problems, including the depletion of resources, accumulation of resources in a few hands (dividing society into rich and poor), and global ecological crises like global warming and ozone layer depletion.

Sustainable Development: This concept means development should take place without damaging the environment, and development in the present should not compromise the needs of future generations.

- **Rio de Janeiro Earth Summit (1992):** More than 100 heads of state met in Brazil to address environmental protection. They signed the Declaration on Global Climatic Change and Biological Diversity.
- **Agenda 21:** Adopted at the Rio Summit, this is an agenda to combat environmental damage, poverty, and disease through global cooperation and shared responsibilities.
- **Resource Conservation:** As Mahatma Gandhi stated, "There is enough for everybody's need and not for any body's greed."

4. Resource Planning in India

Resource planning is crucial in a country like India due to its enormous diversity. Some regions are rich in minerals (e.g., Jharkhand, Madhya Pradesh) but lack infrastructure, while others (e.g., Ladakh) have rich cultural heritage but lack water and minerals. Planning involves three main processes:

- **Identification and Inventory:** Surveying, mapping, and qualitative/quantitative estimation of resources.
- **Planning Structure:** Evolving a structure endowed with appropriate technology, skill, and institutional setup.
- **Matching Plans:** Aligning resource development plans with overall national development plans.

5. Land Resources

Land is a finite asset supporting natural vegetation, wildlife, and economic activities. India possesses a variety of relief features:

- **Plains (43%):** Provide facilities for agriculture and industry.
- **Mountains (30%):** Ensure perennial flow of rivers and support tourism/ecology.

- **Plateaus (27%):** Rich reserves of minerals, fossil fuels, and forests.

6. Land Use Pattern and Degradation

Land use is determined by physical factors (topography, soil) and human factors (population, technology). The forest area in India is significantly lower than the desired 33% outlined in the National Forest Policy (1952).

Causes of Land Degradation:

- **Mining:** Responsible for severe degradation in Jharkhand, Chhattisgarh, Madhya Pradesh, and Odisha.
- **Overgrazing:** A major cause in Gujarat, Rajasthan, Madhya Pradesh, and Maharashtra.
- **Over-irrigation:** Leads to water logging, salinity, and alkalinity in Punjab, Haryana, and Western Uttar Pradesh.
- **Industrial Effluents:** Waste causes water and land pollution in industrial areas.

7. Classification of Soils: Alluvial and Black Soil

Alluvial Soil: The most widely spread soil, covering the northern plains and eastern coastal deltas. Formed by deposits from the Indus, Ganga, and Brahmaputra rivers. It consists of sand, silt, and clay and is very fertile (rich in potash, phosphoric acid, lime).

Black Soil (Regur): Ideal for growing cotton. Found in the Deccan trap (Basalt) region (Maharashtra, Malwa, MP). It is made of lava flows.

- **Age of Alluvial Soil:** Classified into **Bangar** (old alluvial, higher concentration of 'Kankar' nodules) and **Khadar** (new alluvial, more fine particles, more fertile).
- **Features of Black Soil:** Extremely fine clayey material, high moisture-holding capacity, and develops deep cracks in hot weather (helps in aeration).

8. Other Major Soil Types

Red and Yellow Soils: Found in low rainfall areas of the Deccan plateau. They develop a reddish colour due to the diffusion of iron in crystalline rocks and look yellow when hydrated.

Laterite Soil: Results from intense leaching due to heavy rain (tropical climate). It is acidic ($pH < 6.0$) and generally humus-poor. Found in Karnataka, Kerala, and Tamil Nadu. Suitable for tea, coffee, and cashew nut.

Arid Soils: Sandy texture and saline nature. Lacks humus and moisture. The lower horizons contain 'Kankar' nodules (calcium) which restrict water infiltration.

- **Forest Soils:** Found in hilly areas. Texture varies from loamy/silty in valley sides to coarse-grained in upper slopes.

9. Soil Erosion and Conservation

Soil erosion is the removal of soil cover and its washing down. It is caused by natural forces (wind, water) and human activities (deforestation, defective farming).

Types of Erosion:

1. **Gully Erosion:** Running water cuts deep channels (gullies) into clayey soil, creating unfit 'bad land' (e.g., Ravines in Chambal).

2. **Sheet Erosion:** Topsoil is washed away by water flowing as a sheet over large areas.

3. **Wind Erosion:** Wind blows loose soil off flat or sloping land.

- **Contour Ploughing:** Ploughing along contour lines to slow down water flow.
- **Terrace Cultivation:** Cutting steps on slopes to restrict erosion (Western/Central Himalayas).
- **Strip Cropping:** Growing strips of grass between crops to break wind force.
- **Shelter Belts:** Planting rows of trees to stabilize sand dunes and break wind force.